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Can ChatGPT identify predatory biomedical and dental journals? A cross-sectional content analysis

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ABSTRACT

Objectives: To assess whether ChatGPT can help to identify predatory biomedical and dental journals, analyze the content of its responses and compare the frequency of positive and negative indicators provided by ChatGPT concerning predatory and legitimate journals.
Methods: Four-hundred predatory and legitimate biomedical and dental journals were selected from four sources: Beall's list, unsolicited emails, the Web of Science (WOS) journal list and the Directory of Open Access Journals (DOAJ). ChatGPT was asked to determine journal legitimacy. Journals were classified into legitimate or predatory. Pearson's Chi-squared test and logistic regression were conducted. Two machine learning algorithms determined the most influential criteria on the correct classification of journals.
Results: The data were categorized under 10 criteria with the most frequently coded criteria being the transparency of processes and policies. ChatGPT correctly classified predatory and legitimate journals in 92.5 % and 71 % of the sample, respectively. The accuracy of ChatGPT responses was 0.82. ChatGPT also demonstrated a high level of sensitivity (0.93). Additionally, the model exhibited a specificity of 0.71, accurately identifying true negatives. A highly significant association between ChatGPT verdicts and the classification based on known sources was observed ($P < 0.001$). ChatGPT was 30.2 times more likely to correctly classify a predatory journal (95 % confidence interval: 16.9–57.43, p-value: <0.001).
Conclusions: ChatGPT can accurately distinguish predatory and legitimate journals with a high level of accuracy. While some false positive (29 %) and false negative (7.5 %) results were observed, it may be reasonable to harness ChatGPT to assist with the identification of predatory journals.
Clinical significance statement: ChatGPT may effectively distinguish between predatory and legitimate journals, with accuracy rates of 92.5 % and 71 %, respectively. The potential utility of large-scale language models in exposing predatory publications is worthy of further consideration.

1. Introduction

The scientific publishing model is in a state of relative flux due to the advent of open access (OA) publishing. This has contributed to a wider dissemination of scientific knowledge, with journal finances increasingly supported by an article processing charge (APC) paid to the publisher by the author. As such, readers may be able to access articles freely without subscription. Predatory journals have capitalized on this development, exploiting researchers who may be pressured to "publish or perish". This has prompted a proliferation of these predatory journals many of which have evaded detection and leaked into trustworthy databases [1]. This issue is further complicated by the difficulty in pinpointing predatory behavior and the presence of predatory indicators both in presumed legitimate and predatory journals [2]. This confusion has resulted in comparable online engagement with dental articles shared on social media platforms, regardless of whether the articles are published in legitimate or predatory journals [3].

Signals of predatory publishers include cursory or absent peer-

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review; the use of unsolicited emails exhorting publication; the lack of an editorial board, transparent operating procedures and archives; violation of manuscript copyrights; and websites with grammatical errors and poor-quality images [2,4]. The biomedical and dental fields have not been immune with predatory publishing threatening the integrity of the evidence base. Identification of predatory publications is, however, complex involving the use of multiple databases as well as subjective assessment [3]. Furthermore, available checklists such as Think.Check.Submit [5] are useful but lengthy and reliant on subjective assessment. These approaches either involving frameworks (converted to metrics or flowcharts) or lists (whitelists or blacklists) have received criticism being based on diverse and occasionally contradictory criteria with limited validation [6]. More advanced and sensitive approaches to identify these journals are therefore required.

Larger-scale artificial intelligence (AI) models have enabled the introduction of human-like interfaces that receive online human feedback, affording more flexibility in performing and learning intellectual tasks that were formerly the preserve of humans [7]. AI-driven tools can help in website classification by implementing traditional machine-learning as well as deep learning approaches, although these strategies should be applied with caution to avoid the mis-classification of reputable journals as predatory [8,9]. Information on journal website is aggregated through a process called "website scraping" [10]. Key indicators include grammatical and spelling mistakes, different languages on the same page, claims concerning metrics or indexing, and the absence of copyright policy or publishing license. A predefined framework of AI techniques is used to educate the system to detect any significant discrepancies across the publishing venues [11].

ChatGPT, a large-scale language model launched by Open-AI, has shown an exceptional ability in articulating AI-generated responses to queries [12–14]. It is built on large language models (LLMS), which are deep learning algorithms that recognize, summarize, predict and generate text content based on knowledge gained from large databases [15]. ChatGPT has been utilized in healthcare [16,17] and dentistry [18] and may help in scientific research by creating literature summaries and analyzing clinical data sets [19]. Notwithstanding this, its use imposes risks related to biased content due to the type of training datasets [20], allied to legal and ethical concerns related to data privacy and transparency, and the malicious content [20–22]. Moreover, the generation of false scientific content which may appear sound to non-experts is also possible [23].

According to ChatGPT, it can be used to identify predatory journals (Box 1). However, its value in this regard has not been evaluated. Therefore, the aim of this study was to assess whether ChatGPT can help to identify predatory biomedical and dental journals. Furthermore, we aimed to analyze the content of ChatGPT responses and compare the frequency of positive and negative indicators provided by ChatGPT concerning predatory and legitimate biomedical and dental journals.

2. Methods

A total of 400 biomedical and dental predatory and legitimate journals were selected from four sources (Fig. 1). Predatory journals with English titles were randomly selected from Beall's list (https://beallslist.net/) ($n = 100$), and unsolicited emails ($n = 100$) requesting manuscript submission received by the authors over a 4-month period (Jan 1st to May 1st 2023). Legitimate journals with English titles were randomly selected from two additional sources: the Web of Science (WOS) journal list (Science Citation Index Expanded (SCIE) core collection; https://mjl.clarivate.com/home) ($n = 100$), and the Directory of Open Access Journals (DOAJ; https://doaj.org/) ($n = 100$). The sample size per group was pre-determined to ensure equal numbers of biomedical (n= 50) and dental journals (n= 50) from each of the four sources (Beall's list, unsolicited emails, WOS journal list and DOAJ). Biomedical (biomed*, med*, health or clinical) or dental (dent*, oral or stomatology) journals were identified using specific keywords. Journals that could be categorized as both medical and dental were excluded.

The journals titles were crosschecked on ChatGPT (May Version – free access) [24] using the following question: "Is there any sign that ….. is predatory?". A number of prompts were used if ChatGPT provided a generic answer (Supplementary 1).

The journal website was added to the question when ChatGPT incorrectly identified the journal (referring to an incorrect publisher), or when ChatGPT responded: "I'm unable to determine which journal you are referring to… There are many journals with similar title… Please provide me with more details so I can assist you better".

The answers provided by ChatGPT were analyzed in terms of length and content; the latter was based on key criteria and further categorized into positive and negative indicators related to journal legitimacy. These were then ordered into a coding guide. Journals were subsequently classified into those with negative indicators only, positive indicators only or both (Supplementary 2). Based on the answers provided by ChatGPT, a journal was classified as predatory if at least one negative indicator was mentioned, and was classified as legitimate if only positive indicators reported. The coding guide was updated as new criteria emerged. Pilot coding was undertaken on a subset of 50 answers by two authors (DA and SAA) prior to data collection, and disagreements were resolved by joint discussion. The length of answers provided by ChatGPT was recorded. Cohen's Kappa was calculated to assess inter-examiner reliability (DA and SAA) for the classification of the journals [25]. There was excellent agreement between raters ($\kappa = 0.98$).

Descriptive statistics were performed to determine the frequency of positive and negative indicators in the selected journals based on ChatGPT responses. The following metrics were calculated: accuracy, sensitivity, specificity, positive predictive value, negative predictive value, positive likelihood ratio, and negative likelihood ratio. Pearson's Chi-squared test and logistic regression were conducted to assess the ability of ChatGPT to give an accurate indication of journal legitimacy. In addition, a logistic regression model was fitted to examine the effect of the journal source on its legitimacy based on ChatGPT responses.

Two machine learning algorithms were subsequently applied in order to determine the most influential predictors (or criteria) on the correct classification of journals into predatory or legitimate. The random forest model (randForest R package) was initially employed [26]. This model uses bootstrapping to create multiple copies of the original training dataset, fitting a separate decision tree to each copy, followed by combining the results to create a single predictive model.

Box 1
ChatGPT's response regarding its ability to detect predatory journals (May 2023)

Can ChatGPT check whether a journal is predatory or not?

"*Yes, I can certainly help with that. However, it's important to note that identifying predatory journals can be challenging, and there are many potential signs to look out for. I can use my training data to check for some common indicators of predatory publishing practices, but it's always a good idea to conduct your own research and exercise caution before submitting your work to any journal.*"

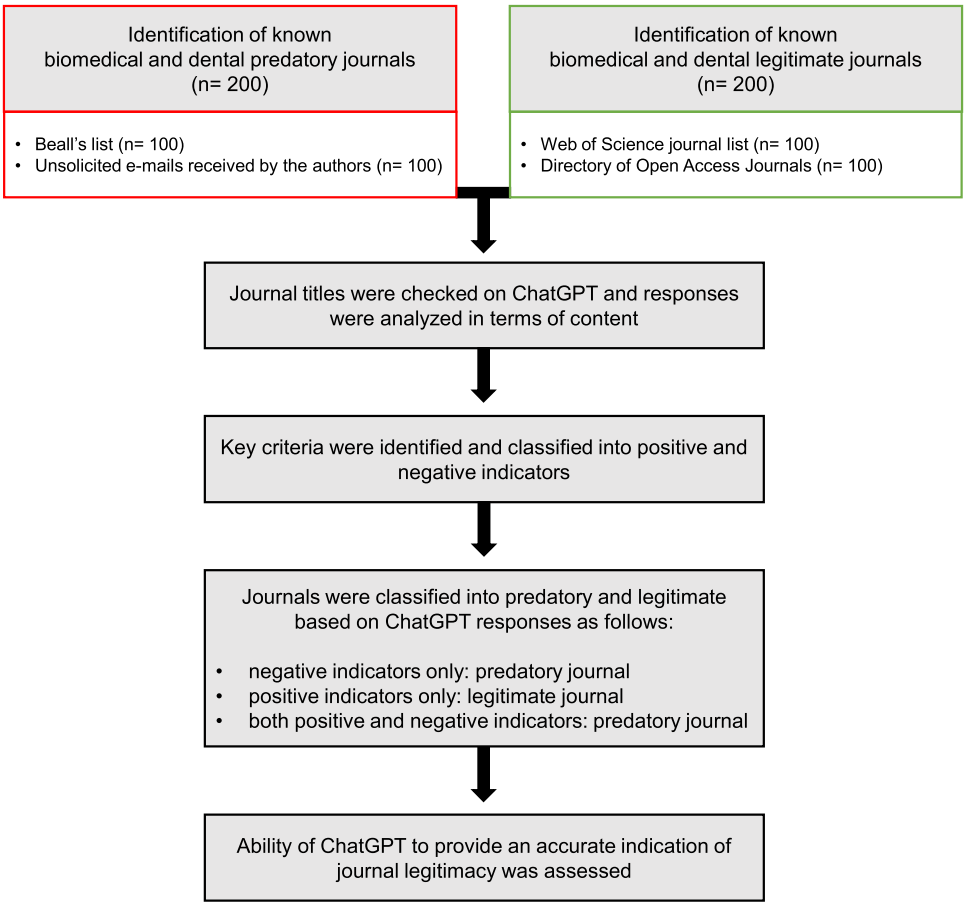


Fig. 1. Summary of methods flowchart.

The selected number of trees was 500 and the number of parameters at the split was $m = 4$; selected after applying the random forest tuning algorithm for the selection of the optimal m. The gradient boosting model [27] (gbm R package) was fitted and updated after assessing the best number of iterations (number of trees to fit) using different cross-validation folds, interaction depth, and shrinkage (learning rate or step-size reduction) parameter values. Receiver operating characteristic (ROC) plots and area under the curve (AUC) were calculated for both random forest and gradient boosting.

Statistical significance was set at a p-value of ≤ 0.05. All statistical analysis were undertaken using the R statistical package (version 4.3.1) (R Foundation for Statistical Computing, Vienna, Austria).

3. Results

A total of 400 journals were included from two predatory and two legitimate sources with equal numbers of biomedical and dental journals. The mean word count of ChatGPT responses for predatory journals ranged from 215.8 to 239.4 and 163.3 to 182.1 words in biomedical and dental journals, respectively. Furthermore, the mean word count for legitimate journals ranged from 231.9 to 256 and 233.1 to 259.9 words in biomedical and dental journals, respectively.

Table 1
Frequencies of negative and positive indicators identified in predatory and legitimate journals.

	Predatory journals ($n = 200$)		Legitimate journals ($n = 200$)	
	Frequency of negative indicators	Frequency of positive indicators	Frequency of negative indicators	Frequency of positive indicators
Criteria 1: Transparency of processes and policies	$n = 467$	$n = 44$	$n = 107$	$n = 172$
Criteria 2: Indexing, abstracting and archiving status and identification	$n = 109$	$n = 26$	$n = 33$	$n = 107$
Criteria 3: Publication fees	$n = 43$	$n = 23$	$n = 6$	$n = 2$
Criteria 4: Peer review and publication practices of the journal	$n = 13$	$n = 4$	$n = 0$	$n = 180$
Criteria 5: Scope of the journal	$n = 54$	$n = 4$	$n = 0$	$n = 3$
Criteria 6: Publisher and publishing practices	$n = 45$	$n = 21$	$n = 8$	$n = 278$
Criteria 7: Website quality	$n = 280$	$n = 15$	$n = 7$	$n = 92$
Criteria 8: Reputation of the journal and duration of its existence	$n = 10$	$n = 3$	$n = 1$	$n = 106$
Criteria 9: Metrics	$n = 14$	$n = 0$	$n = 28$	$n = 12$
Criteria 10: Aggressive solicitations	$n = 3$	$n = 0$	$n = 0$	$n = 2$

*Detailed list of indicators presented in Supplementary 3.

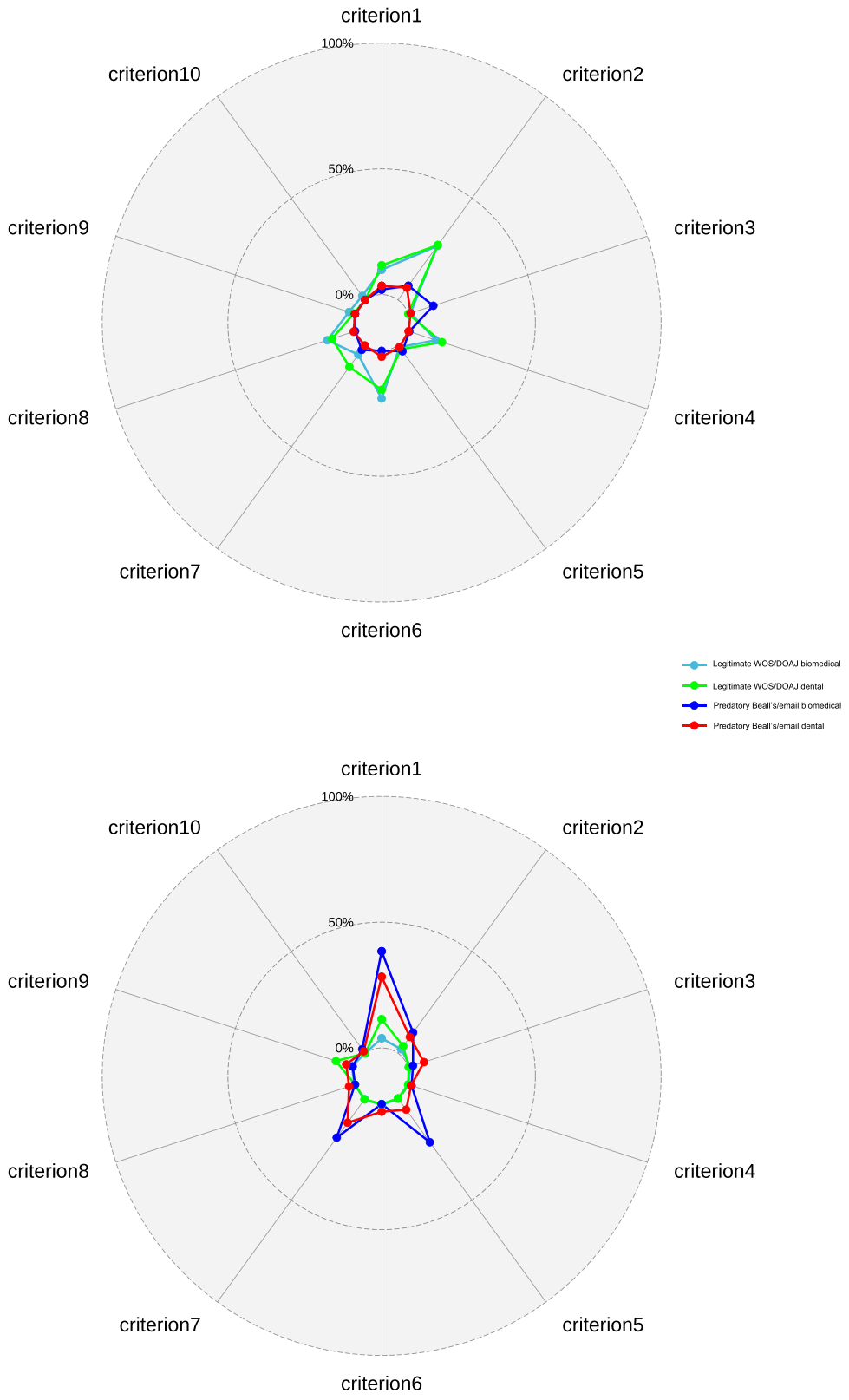


Fig. 2. Radial chart depicting the frequency of positive indicators (top) and negative indicators (below).
DOAJ: Directory of Open Access Journals; WOS: Web of Science.

The data were categorized under 10 criteria with the most frequently coded criteria being the transparency of processes and policies, and website quality (Table 1; Supplementary 3). In terms of "the transparency of processes and policies" concerns identified by ChatGPT for predatory journals were mostly related to ambiguous information with regard to editorial board members (93 % of biomedical journals and 76 % of dental journals), and the peer review process (93 % of biomedical journals and 79 % of dental journals) (Table 1; Supplementary 3).

Table 2
Overall judgement of the included journals ($n = 400$) based on GhatGPT responses.

Journal classification based on known sources	Scope of journal	Source	Frequency of journals with different types of indicators based on ChatGPT responses		
			Positive indicators only	Both positive and negative indicators	Negative indicators only
Predatory journals	Biomedical	Beall's list ($n = 50$)	$n = 4$	$n = 21$	$n = 25$
		Unsolicited emails ($n = 50$)	$n = 1$	$n = 3$	$n = 46$
	Dental	Beall's list ($n = 50$)	$n = 4$	$n = 5$	$n = 41$
		Unsolicited emails ($n = 50$)	$n = 6$	$n = 5$	$n = 39$
Legitimate journals	Biomedical	WOS ($n = 50$)	$n = 50$	$n = 0$	$n = 0$
		DOAJ ($n = 50$)	$n = 28$	$n = 21$	$n = 1$
	Dental	WOS ($n = 50$)	$n = 39$	$n = 9$	$n = 2$
		DOAJ ($n = 50$)	$n = 25$	$n = 22$	$n = 3$

DOAJ: Directory of Open Access Journals; WOS: Web of Science.

Table 3
Confusion matrix for journal classification based on known sources vs ChatGPT responses.

		Classification based on known sources		p-value*
		Predatory ($n = 200$)	Legitimate ($n = 200$)	
Classification based on ChatGPT responses	Predatory	$n = 185$ (92.5 %)	$n = 58$ (29 %)	<0.001
	Legitimate	$n = 15$ (7.5 %)	$n = 142$ (71 %)	

* Pearson's Chi-squared test.
Accuracy: 0.8175 (95 % confidence interval (CI): 0.78, 0.85).
Sensitivity: 0.925 (95 % CI: 0.879, 0.957).
Specificity: 0.71 (95 % CI: 0.642, 0.772).
Positive predictive value: 0.761 (95 % CI: 0.703, 0.813).
Negative predictive value: 0.904 (95 % 0.847, 0.946 %).
Positive likelihood ratio (LR+): 3.1897.
Negative likelihood ratio (LR-): 0.1056.

However, legitimate journals also had negative indicators mostly related to the limited information available about the peer review process (17 % of biomedical journals and 28 % of dental journals).

The concerns noted by ChatGPT for predatory journals related to "website quality" were mainly associated with poorly designed, low-quality websites (86 % of biomedical journals and 50 % of dental journals) (Table 1; Supplementary 3). In terms of legitimate journals, ChatGPT indicated a well-designed and professional website (12 % of biomedical journals and 37 % of dental journals).

Both positive and negative indicators were reported for predatory and legitimate journals (Fig. 2; Table 1). However, the frequency of negative indicators in predatory journals exceeded positive indicators for all criteria; and were 7 times more prevalent overall (1038 vs 140) (Table 1). Regarding legitimate journals, positive indicators outstripped negative indicators for all criteria except in criteria "metrics" (12 vs 28), and criteria related to publication fees (2 vs 6). In legitimate journals, positive indicators were 5 times more common than negative indicators overall (954 vs 190) (Table 1).

In terms of scope of journal, ChatGPT cited negative indicators for the majority of both biomedical (95 %) and dental (90 %) predatory journals (Table 2). Very limited numbers of predatory journals in biomedical (5 %) and dental (10 %) groups contained positive indictors only in the ChatGPT answers (Table 2). In terms of legitimate journals, 78 % and 64 % of biomedical and dental journals had positive indicators only in the ChatGPT answers, respectively (Table 2).

3.1. Accuracy of ChatGPT responses

ChatGPT correctly classified predatory and legitimate journals in 92.5 % and 71 % of the sample, respectively (Table 3). The accuracy of ChatGPT responses was 0.82 (95 % confidence interval (CI): 0.78, 0.85). ChatGPT demonstrated a high level of sensitivity, with a value of 0.93 (95 % CI: 0.88, 0.96), indicating a strong ability to accurately detect predatory journals. Additionally, ChatGPT exhibited a specificity of 0.71 (95 % CI: 0.64, 0.77), accurately identifying true negatives (legitimate journals). Positive and negative predictive values were 0.76 and 0.9, respectively.

A highly significant association between ChatGPT verdicts and the classification based on known sources was identified (Table 4). ChatGPT was 30.2 times more likely to classify a predatory journal as predatory compared to legitimate (95 % CI: 16.9–57.43, p-value: <0.001) (Table 4). Based on the findings of the logistic regression, the scope of journal (biomedical or dental) did not have an effect on its classification, with an odds ratio for dental journals of 1.21, (95 % CI: 0.81 – 1.81, p-value: 0.36).

Based on the findings of the logistic regression, ChatGPT performance was affected by the journal source (Fig. 3). Journals identified from predatory sources including Beall's list (0.92) and from unsolicited emails received by the authors (0.91) had higher probabilities of being classified as predatory based on ChatGPT answers compared to journals identified from legitimate sources including DOAJ (0.47) and WOS (0.11) (Fig. 3).

The influence bar plot from gradient boosting and the dot plot from the random forest algorithms ranked the most influential predictors for classifying journals as predatory or legitimate (Fig. 4; Supplementary 4 and 5). Slight variations in the influence of various criteria were noted due to differences in implemented approaches. Overall, the most influential negative and positive predictors for correct journal classification were related to criterion 1 (transparency of processes and policies) and criterion 6 (publisher and publishing practices), respectively. . In the ROC plots, the area under the curve (AUC) metric indicates excellent accuracy of the random forest and gradient boosting (Figs. 5, 6).

Table 4
Logistic model showing the association between journal classification based on ChatGPT responses and known sources.

	Journal classification based on ChatGPT responses		
Predictors	*Odds ratios*	*95 % confidence interval*	*p-value*
Legitimate	*Reference*		
Predatory	30.20	16.90 – 57.43	<0.001
Observations	400		

4. Discussion

The current study highlights the ability of ChatGPT to correctly identify predatory and legitimate journals in 92.5 % and 71 % of the sample, respectively. The majority of biomedical (95 %) and dental (90 %) predatory journals had negative indicators regarding legitimacy. In terms of legitimate journals, positive indicators were present in 78 % and 64 % of biomedical and dental journals, respectively. While these findings are promising, the presence of both false positives (29 %) and false negatives (7.5 %), along with the existence of predatory and legitimate indicators in legitimate and predatory dental journals, respectively, underscore the ongoing challenge of pinpointing predatory journals.

The tools used to identify predatory journals draw parallels with the diagnostic criteria employed in the context of medical diseases [28]. The detrimental consequences of their inaccuracy, whether resulting in reputational damage through publishing in predatory journals, or ultimately incorrect diagnosis and patient care are highly consequential [28]. Arguably, therefore, unless these tools are entirely accurate, they may inadvertently risk harm rather than serving as a solution [28,29]. It is important that ChatGPT is able to detect predatory journals, because researchers and clinicians can use the resource to search for medical information. At this stage, these tools should not replace researchers' own judgement and expertise. However, it is worth noting that AI improves over time. Therefore, our findings are likely to be bettered with future iterations and associated refinement. Nevertheless, the data used within ChatGPT to inform its responses remains opaque. Moreover, in one study, ChatGPT provided fake references in answering medical questions [30]. Therefore, authors must exercise caution and verify the answers provided.

The lack of clear guideline to identify predatory journals has facilitated the proliferation of these journals [4]. These journals are dynamic, adapting to conceal pernicious practices, aiming to emulate legitimate journals, and finding a way to infiltrate trustworthy databases [31] complicating identification by effectively presenting a "moving target" [32]. Furthermore, distinguishing between predatory and legitimate journals is complicated. In one study focused on dental articles published in legitimate and predatory journals, both journal types shared common predatory indicators (e.g. lack of manuscript withdrawal/retraction policy and lack of digital archiving), although the median score was higher in predatory journals [3]. Likewise, some predatory characteristics are also associated with presumed legitimate journals in oral health [33]. Similarly, a number of positive policies (e.g. manuscript preparation guidance or plagiarism policy) and negative characteristics (e.g. no mentioning retraction policy, no statement about copyright retention) were found in both presumed legitimate and predatory publishing outputs in biomedical journals, shrouding the discrimination according to their legitimacy [2]. Additionally, legitimate journals exhibit some deficiencies [4]. Furthermore, some criteria such as peer review are difficult to assess for all journals [34].

The issue of proliferation of predatory journals is further compounded by their infiltration into legitimate databases, their presence on social media platforms and targeting of authors by email [3,31,35]. For instance, one researcher reported receiving in excess of 1800 emails requesting manuscript submission in a single year [35]. These emails are often sent from new predatory journals with short publication history,

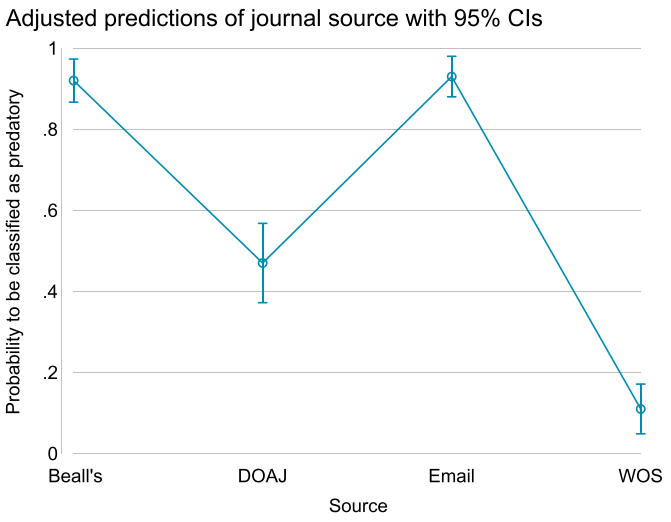


Fig. 3. The probability of journal classification as predatory based on ChatGPT responses considering the source of the journal.
CI: confidence interval; DOAJ: Directory of Open Access Journals; WOS: Web of Science.

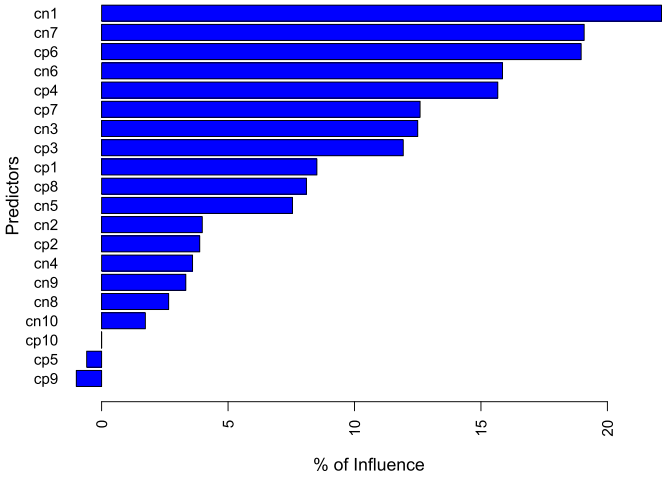


Fig. 4. Bar plot showing the influence of various criteria on the model accuracy (gradient boosting).
c: criteria; n: negative; p: positive.

employing flattering language to attract authors [35,36]. In one study, it was found that only 76 % of emails received by a single author over the course of a year were blacklisted [36]. There is therefore a need for live archives that track journals sending unsolicited emails. However, it is crucial to exercise caution due to the potential harm that may be caused by competing interests.

In the current study, the comparison of ChatGPT responses was made with existing resources, a pragmatic decision considering the widespread reliance on these lists. This comparison was deemed timely given that novice researchers may turn to AI to verify journal legitimacy due its ease of use and availability, in addition to the recognized limitations of available lists. For instance, Beall's list has been criticized for being outdated and producing false negative results due to bias in inclusion of open access journals [37]. Furthermore, it has been criticized for its lack of transparency and detailed explanation [38]. Conversely, ChatGPT provides detailed explanations, although the accuracy of these explanations was not verified in the current study. Additionally, other blacklists such as Cabell's has faced criticism due to inaccessibility, and was therefore, not used in the current study [39]. Limitations of

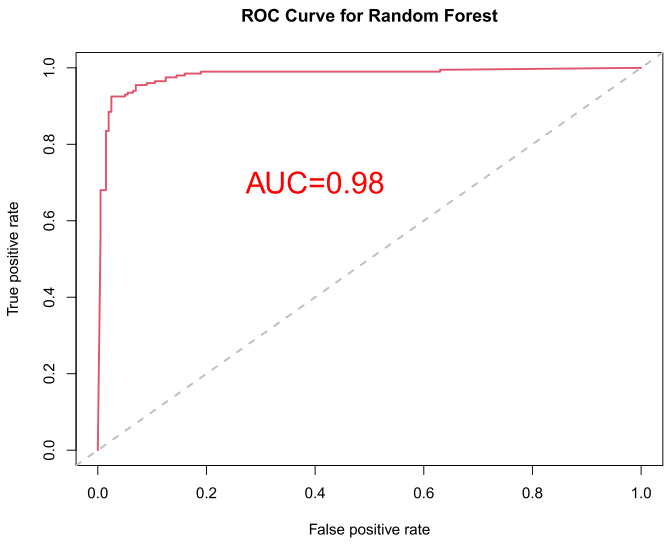


Fig. 5. ROC plot for the random forest (AUC= 0.98).

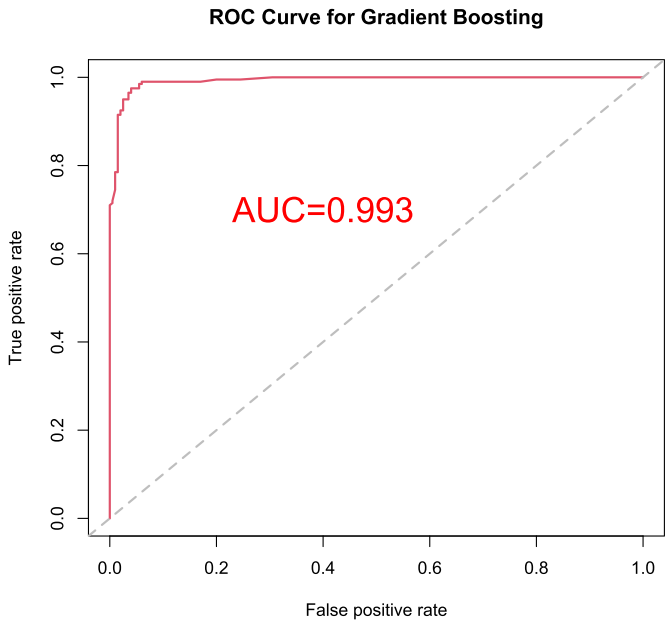


Fig. 6. ROC curve for gradient boosting (AUC= 0.993).

available blacklists and whitelists are clear, with some journals simultaneously found in both lists [4]. The use of checklists can be perplexing with over 90 being available, many of which have limitations including a lack of a clear and definitive threshold and evidence-based support [40]. Therefore, we require a responsive and current system to address the dynamic nature of predatory publishing.

ChatGPT answers provided positive and negative indicators, without assigning a binary assessment of predatory or legitimacy. The authors of a previous study argued that the quality of academic publishing exists along a continuum rather than a binary classification [41]. A score is assigned using a credit-like rating system, taking into account both positive and negative characteristics [42,43] with black and white at the opposing ends of the spectrum, and grey in the middle [41]. The lack of appreciation of this grey zone, can result in erroneous classification of journals as either predatory or legitimate [41]. Therefore, the concept of binary classification might be questionable.

It is clear that any efforts to identify predatory journals must incorporate continuous updates, free accessibility, and independent verification and approval by publishing organizations such as the Committee on Publication Ethics (COPE) [41]. A number of approaches have been suggested including the establishment of online forums for discussions on journal quality [34]. Furthermore, journals are encouraged to adopt transparent peer review practices urging authors to publicly share their experiences with peer review processes [34]. Online tools have been utilized to summarize text posts on social media (Reddit) about predatory journals [44]. This facilitates timely access to information related to current practices of predatory journals [44]. Another proposed solution includes the development of software designed to detect predatory journal websites, akin to systems used for identifying phishing websites [45]. Furthermore, the idea of creating a website, similar to an online repository, for researchers to report questionable journals has been suggested [44]. A similar concept would be useful for maintaining record of journals sending unsolicited emails. Institutions and organizations also have a role to play by providing clear guidance for the detection of predatory journals. Furthermore, the active involvement and collaboration of librarians can play a pivotal role in preventing more naïve researchers from publishing in these journals [46].

Freely-available large-scale language systems could serve as a valuable tool for researchers, publishers, and academic institutions in their efforts to combat predatory publishing. However, future updates of ChatGPT should be focused on reducing false positives and false negatives to enhance the system's overall performance. At this stage it is unclear if predatory journals can also deceive AI and contaminate its content; therefore, enhanced transparency on how the system operates would be beneficial. Additionally, the generalizability of the system should be examined, as ChatGPT may perform differently when applied to journals in various disciplines. It is worth noting that a cross-sectional evaluation was undertaken. We were therefore unable to assess the consistency of answers over time nor indeed the responsiveness of ChatGPT to journal practices.

5. Conclusions

The current study demonstrated the potential of ChatGPT to effectively identify predatory and legitimate journals, with respective accuracy rates of 92.5 % and 71 % and a highly significant association between ChatGPT verdicts and the classification based on known sources. While these findings offer promise, the presence of false positive and false negative findings underscores the persistent challenge of exposing predatory journals. Notwithstanding this, large-scale language models may already be a useful adjunct in journal classification.

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CRediT authorship contribution statement

Dalya Al-Moghrabi: Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Sarah Abu Arqub:** Writing – review & editing, Investigation, Data curation. **Michael P Maroulakos:** Writing – original draft, Investigation, Data curation. **Nikolaos Pandis:** Writing – review & editing, Methodology, Investigation, Formal analysis. **Padhraig S Fleming:** Writing – review & editing, Methodology, Investigation, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.jdent.2024.104840.

References

[1] M. Somoza-Fernández, J.M. Rodríguez-Gairín, C. Urbano, Presence of alleged predatory journals in bibliographic databases: analysis of Beall's list, Prof. Inf. 25 (5) (2016).

[2] L. Shamseer, D. Moher, O. Maduekwe, L. Turner, V. Barbour, R. Burch, J. Clark, J. Galipeau, J. Roberts, B.J. Shea, Potential predatory and legitimate biomedical journals: can you tell the difference? A cross-sectional comparison, BMC Med. 15 (1) (2017) 1–14.

[3] D. Al-Moghrabi, R.S. Albishri, R.D. Alshehri, S.A. Arqub, A. Alkadhimi, P. S. Fleming, An analysis of dental articles in predatory journals and associated online engagement, J. Dent. 129 (2023) 104385.

[4] A. Grudniewicz, D. Moher, K.D. Cobey, G.L. Bryson, S. Cukier, K. Allen, C. Ardern, L. Balcom, T. Barros, M. Berger, Predatory Journals: No Definition, No Defence, Nature Publishing Group, 2019.

[5] https://thinkchecksubmit.org/ 2024.

[6] T.F. Frandsen, How can a questionable journal be identified: frameworks and checklists, Learn. Publ. 32 (3) (2019) 221–226.

[7] D.Q. Wang, L.Y. Feng, J.G. Ye, J.G. Zou, Y.F. Zheng, Accelerating the integration of ChatGPT and other large-scale AI models into biomedical research and healthcare, MedComm Future Med. 2 (2) (2023) e43.

[8] J.A. Teixeira da Silva, P. Tsigaris, S. Moussa, Can AI detect predatory journals? The case of FT50 journals, Case FT50 J. (2023), https://doi.org/10.2139/ssrn.4391108.

[9] M. Andoohgin Shahri, M.D. Jazi, G. Borchardt, M. Dadkhah, Detecting hijacked journals by using classification algorithms, Sci. Eng. Ethics 24 (2018) 655–668.

[10] L.X. Chen, K.S. Wong, C.H. Liao, S.M. Yuan, Predatory journal classification using machine learning, in: Proceedings of the 3rd IEEE International Conference on Knowledge Innovation and Invention (ICKII), IEEE, 2020, pp. 193–196.

[11] K. Kowsari, K.J. Meimandi, M. Heidarysafa, S. Mendu, L. Barnes, D. Brown, Text classification algorithms: a survey, Information 10 (4) (2019) 150, arXiv preprint arXiv:1904.08067.

[12] Ö. Aydın, E. Karaarslan, OpenAI ChatGPT generated literature review: digital Twin in Healthcare, Available at SSRN 4308687 (2022).

[13] C.A. Gao, F.M. Howard, N.S. Markov, E.C. Dyer, S. Ramesh, Y. Luo, A.T. Pearson, Comparing scientific abstracts generated by ChatGPT to original abstracts using an artificial intelligence output detector, plagiarism detector, and blinded human reviewers, BioRxiv (2022) 2022.12. 23.521610.

[14] X. Zhai, ChatGPT user experience: implications for education, Available at SSRN 4312418 (2022).

[15] G. Eysenbach, The role of ChatGPT, generative language models, and artificial intelligence in medical education: a conversation with ChatGPT and a call for papers, JMIR Medical Education 9 (1) (2023) e46885.

[16] T.P. Nagarhalli, V. Vaze, N. Rana, A review of current trends in the development of chatbot systems, in: Proceedings of the 6th International conference on advanced computing and communication systems (ICACCS), IEEE, 2020, pp. 706–710.

[17] M. Sallam, ChatGPT utility in healthcare education, research, and practice: systematic review on the promising perspectives and valid concerns. Healthcare, MDPI, 2023, p. 887.

[18] ADA releases report on AI in dentistry - American dental association 2024 https://www.ada.org/publications/ada-news/2023/february/ada-releases-report-on-ai-in-dentistry (Last accessed: 22/09/2023).

[19] M. Sallam, The utility of ChatGPT as an example of large language models in healthcare education, research and practice: systematic review on the future perspectives and potential limitations, medRxiv (2023) 2023.02. 19.23286155.

[20] B. Marchandot, K. Matsushita, A. Carmona, A. Trimaille, O. Morel, ChatGPT: the next frontier in academic writing for cardiologists or a pandora's box of ethical dilemmas, Eur. Heart J. Open 3 (2) (2023) oead007.

[21] J. Kleesiek, Y. Wu, G. Stiglic, J. Egger, J. Bian, An opinion on ChatGPT in health care—written by humans only, Soc Nuclear Med (2023) 701–703.

[22] G. Kendall, J.A. Teixeira da Silva, Risks of Abuse of Large Language Models, Like ChatGPT, in Scientific publishing: Authorship, Predatory publishing, and Paper Mills, Learned Publishing, 2024.

[23] Y. Shen, L. Heacock, J. Elias, K.D. Hentel, B. Reig, G. Shih, L. Moy, ChatGPT and Other Large Language Models Are Double-Edged Swords, Radiological Society of North America, 2023 e230163.

[24] C.M.v 2024 https://help.openai.com/en/articles/6825453-chatgpt-release-notes.

[25] A.U.h.d.n. DATAtab Team (2023) DATAtab: online Statistics Calculator. DATAtab e.U. Graz.

[26] L. Breiman, Random forests, Mach. Learn. 45 (2001) 5–32.

[27] G. Ridgeway, The state of boosting, Comput. Sci. Stat. 31 (1999) 172–181.

[28] J.A. Teixeira da Silva, T. Daly, The diagnostic accuracy of AI-based predatory journal detectors: an analogy to diagnosis, Diagnosis 10 (2023), https://doi.org/10.1515/dx-2023-0039.

[29] N. Naik, B. Hameed, D.K. Shetty, D. Swain, M. Shah, R. Paul, K. Aggarwal, S. Ibrahim, V. Patil, K. Smriti, Legal and ethical consideration in artificial intelligence in healthcare: who takes responsibility? Front. Surg. 9 (2022) 266.

[30] J. Gravel, M. D'Amours-Gravel, E. Osmanlliu, Learning to fake it: limited responses and fabricated references provided by ChatGPT for medical questions, Mayo Clin. Proc. Digit. Health 1 (3) (2023) 226–234.

[31] A. Manca, D. Moher, L. Cugusi, Z. Dvir, F. Deriu, How predatory journals leak into PubMed, CMAJ 190 (35) (2018) E1042–E1045.

[32] J.A.T. da Silva, G. Kendall, Academia should stop using Beall's Lists and review their use in previous studies, Cent. Asian J. Med. Hypotheses Ethics 4 (1) (2023) 39–47.

[33] Z. Huseynova, N. Pandis, C.M. Faggion Jr., Presumed predatory journals are abundant in oral health, J. Evid. Based Dent. Pract. 21 (2) (2021) 101539.

[34] B.L. Tang, Are there accurate and legitimate ways to machine-quantify predatoriness, or an urgent need for an automated online tool? Account Res. (2023) 1–6, https://doi.org/10.1080/08989621.2023.2253425.

[35] F.S. de Oliveira Sousa, P. Nadanovsky, I.M. Dhyppolito, A.P.P. Dos Santos, One year of unsolicited e-mails: the modus operandi of predatory journals and publishers, J. Dent. 109 (2021) 103618.

[36] C. Livas, K. Delli, "Dear doctor, greetings of the day!": a 1-year observational study of presumed predatory journal invitations, Prog. Orthod. 24 (1) (2023) 21.

[37] F. Krawczyk, E. Kulczycki, How is open access accused of being predatory? The impact of Beall's lists of predatory journals on academic publishing, The Journal of Academic Librarianship 47 (2) (2021) 102271.

[38] S.G. Kimotho, The storm around Beall's List: a review of issues raised by Beall's critics over his criteria of identifying predatory journals and publishers, Afr. Res. Rev. 13 (2) (2019) 1–11.

[39] X. Chen, Beall's list and Cabell's blacklist: a comparison of two lists of predatory OA journals, Ser. Rev. 45 (4) (2019) 219–226.

[40] S. Cukier, L. Helal, D.B. Rice, J. Pupkaite, N. Ahmadzai, M. Wilson, B. Skidmore, M. M. Lalu, D. Moher, Checklists to detect potential predatory biomedical journals: a systematic review, BMC Med. 18 (2020) 1–20.

[41] Y. Yamada, J.A. Teixeira da Silva, A psychological perspective towards understanding the objective and subjective gray zones in predatory publishing, Qual. Quant. 56 (6) (2022) 4075–4087.

[42] J.T. da Silva, Predatory publishing: a quantitative assessment, the predatory score, Asian Australas. J. Plant Sci. Biotechnol. 7 (Special Issue 1) (2013) 21–34.

[43] J.A. Teixeira da Silva, D.J. Dunleavy, M. Moradzadeh, J. Eykens, A credit-like rating system to determine the legitimacy of scientific journals and publishers, Scientometrics 126 (10) (2021) 8589–8616.

[44] M. Dadkhah, F. Rahimnia, M.H. Oermann, PedCheck: a dashboard for analyzing social media posts about predatory journals, Ser. Rev. 47 (3–4) (2021) 224–230.

[45] M. Dadkhah, Predatory journals/publishers are not sole questionable matter in open access scholarly publishing, they are part of problem, BMJ (2015), https://doi.org/10.1136/bmj.h2470.

[46] J.A. Teixeira da Silva, Academic librarians and their role in disseminating accurate knowledge and information about the gray zone in predatory publishing, New Rev. Acad. Librariansh. 28 (4) (2022) 383–405.